Lab-13 Ex7 - Nested Structure

Ungraded exercise means that this is not counted to the course total. However, we recommend you complete this as a revision of the final examination. Please be reminded that the final examination is a written examination. Writing a program on a piece of paper is different from typing code in a computer.

Consider the following type definitions for representing a Line:

```
typedef struct line {

  struct point {

    double x, y;

  } P1;

  struct point P2;

} Line;
```

The `struct point` is declared within the `struct line` in a nested manner.

Write a C program to read coordinates of two lines one by one and check whether they are in parallel or not.

Two lines are said to be in parallel (coincident lines are considered as parallel as well) if the difference between their slope values is very close to zero. Note that we should not compare double with zero using equality operator (= =). Instead, we can check whether the slope difference is within -0.001 and 0.001.

User Input:

- The first line contains four numbers, indicating the x and y coordinates of line 1's two end points.
- Then, the x and y coordinates of line 2 is supplied.

Program Output:

- Output a line "parallel" or "not parallel" depending on whether the two lines are in parallel or not.

Sample Run (inputs are in bold and italic)

1 0 3.5 2.5

1.5 -1.5 3.5 0.5

```
parallel
```